RESEARCH REPORT · VITKI

Fully tested finding

Runestones gather in family groups by the grave fields – neighbouring stones name the same personal names eight times as often

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ABSTRACT

Background and aim. Where runestones were raised – by grave fields, roads and bridges, or farm boundaries – has long been discussed. The question is whether the stones form groups, what a group's stones share and where groups stand, with settlement density held constant.
Material. Runestones in place according to Rundata: 458 in Uppland and 168 in Södermanland (pattern and names), 167 in Östergötland, Västergötland and Småland (replication) and 657 in all Swedish provinces (grave fields, from the Swedish National Heritage Board's register).
Method. Four pre-registered studies: a point-pattern test against settlement of varying density, replicated on new stones; near against distant pairs of stones; a logistic regression for grave fields, group against lone stone; and a permutation of style groups within the district. The two main tests passed a fixed series of sensitivity analyses under a locked addendum plan; the fourth, a follow-up, has not.
Results. The stones lie closer together than chance gives at 100–1,000 metres in Uppland and Södermanland, and in Götaland ($p = 0.004$). Pairs of stones within 500 metres share a personal name in 17.6 percent of cases, against 2.2 percent at 2–10 kilometres: 7.9 times as often. 82 percent of grouped stones have a grave field within 500 metres against 55 percent of lone ones; odds ratio 2.38. A group spans on average 1.58 style groups against 1.93 for as many stones from the same district. For roads, burial mounds and boundary markers no difference is shown; with density held constant, place names in Husby and -tuna lie somewhat less often near the groups and a medieval church somewhat more often (both indications).
Conclusion. Runestones often stand in groups that belong to a family: the same personal names, by the family's grave field and, by a follow-up not yet sensitivity-tested, raised within one generation. The findings rest largely on Uppland.

Keywords: *runestones, Viking Age, grave fields, family, stone groups, spatial statistics, point patterns, pre-registration*

HIGHLIGHTS

- Runestones often stand in groups, and the stones in a group share personal names – eight times as often as stones further apart.
- The groups stand by the grave fields more often than lone stones – which fits the family's monument by the family's burial place. Whether the lone stones stood by roads or boundaries the registers cannot show well. A follow-up, not yet sensitivity-tested, suggests that a group was raised within one generation.
- The pattern is clearest in Uppland, where the stones still stand; that is also where it is surest.

1. Introduction

Gunnar Ekholm (1950) argued, against earlier scholarship, that runestones stood mainly by roads and waterways: many stand by grave fields because the grave fields lay by the roads, and the lone stones are mostly road monuments, not remnants of ploughed-out grave fields. Torun Zachrisson (1998), studying Uppland and Gästrikland, related runestones to farm, boundary and grave field and proposed that stones marked the farm's boundaries, often where the road entered the farm's land. Lydia Klos (2009) argued that most runestones were raised at grave fields; Shane McLeod (2014) holds that this overstates the grave fields at the expense of other places, such as roads. Birgit Sawyer (2000) read the stones as expressions of family and inheritance, and groups such as Jarlabanke's at Täby and Vallentuna have long been seen as a lineage's monuments.

What is missing is a measure on the whole material: do the stones form groups beyond chance with settlement density held constant, do a group's stones belong together through people or through roads and bridges, and do groups stand elsewhere than lone stones? This study measures that in four pre-registered steps.

2. Pre-registration: hypotheses and plans

Each plan was committed to version control before its analysis was run; SHA-256 checksums below, files under Data and code.

1. **The spacing** (28 October 2026, `9e7f6be50cd5b932…` , addendum `663cae7ff05a94e2…` ; replication 31 October, `e1fb9abc5cf9ec88…`): do stones in place stand closer together than chance gives at 100–1,000 metres, with settlement density constant? Tested in Uppland and Södermanland, then in Östergötland, Västergötland and Småland.
2. **The neighbouring stones** (29 October, `dae9035c05fe5ba3…`): do near pairs (25–500 metres) more often share a personal name, carver, style group or message than pairs 2–10 kilometres apart?

Six tests, Holm-corrected.

3. **The family's place** (29 October, `68361374fa670187…`): do stones in a group (another stone 25–500 metres away) stand more often than lone stones (nearest stone over 1 kilometre away) by grave fields, burial mounds, roads, boundary markers, medieval churches and certain place names, and do they differ in text, height and signing? Two more tests look within the groups (time depth; Christian beside non-Christian stones). Fifteen tests, Holm-corrected.
4. **One generation** (29 October, `17798b75004cd30c…`): do groups span fewer style groups than as many stones drawn from the same district?
5. **The addendum plan** (31 October, `ec4a17c59132e058…`): the project's fixed series of sensitivity analyses for steps 2 and 3, written after the four results were known but before the sensitivity analyses were run; it adds no outcomes.

Step 4 grew out of step 3, where a one-sided test asked only whether groups have *more* style groups than chance gives; the result went strongly the other way. The new plan tested the opposite with a stricter comparison (within the district, not the province) on the same stones: a pre-registered follow-up, not tested on independent data. In step 1 the method was corrected after the first run (4.6), so it was tested unchanged, under a new locked plan, on stones of Östergötland, Västergötland and Småland that no spacing analysis had used; that test carries step 1.

3. Material

- **Sources.** The Scandinavian Runic-text Database (Rundata), 2014 version with RUNDATA.xls updated 2018, frozen for the analyses on 28 October 2026. Grave fields, burial mounds, roads and boundary markers from the Swedish National Heritage Board's register of ancient monuments; place names from OpenStreetMap; medieval churches (built before 1350) from Wikidata; moved stones also from the Board's register Runor (2020 edition).
- **The stones.** Only runestones that according to Rundata stand at their original place with a known location; moved and lost stones are excluded, since their place does not show where they were raised. Steps 1 and 2: Uppland and Södermanland (458 and 168 stones; 420 and 149 sites, stones within 25 metres counting as one). Replication: 167 stones in Östergötland (59), Västergötland (56) and Småland (52), the only other provinces with enough stones in place. Step 3: all Swedish provinces, 185 stones in a group and 472 lone ones.
- **The personal names** are the sponsors' and the dead's names, read by rules. The 20 most common names are not counted, so that a pair does not share a name merely because it was common.
- **Dating** by ornamental style (Gräslund 2006); a style group covers 25–50 years.
- **The carver.** The carver Rundata gives as secure; stones with an uncertain carver or several carvers count as without a known carver.

4. Method

Statistical terms are explained in the glossary, https://vitkirunor.se/en/forskningsrapporter/begrepp.

4.1 The point pattern. Ripley's L-function counts how many neighbours the stones have within a given distance. It is compared with an inhomogeneous Poisson pattern: points placed independently, but more densely where the settled land has more stones. That density comes from kernel smoothing, a moving average over the map whose bandwidth (10 kilometres) sets its width, with a floor of 5 percent of the mean density. A DCLF test (Diggle–Cressie–Loosmore–Ford) sums the departure from that pattern over 100–1,000 metres into one p-value, against 999 simulations (Baddeley et al. 2015). Uppland and Södermanland separately, Holm-corrected. The replication applies the same method to Götaland's three provinces as one pattern and succeeds if p is below 0.05, the pattern is clustered and the direction holds in both halves.

4.2 The neighbouring stones. For each trait: the share of near pairs sharing it divided by the share of comparison pairs sharing it, with p from 9,999 permutations of the stones' traits within the province, sites fixed.

4.3 The grave fields. Logistic regression: grave field within 500 metres ~ in a group + province + dating period + local density of stones, so that a dense district does not itself count as a pattern.

4.4 One generation. For each group with at least two style-dated stones: the number of different style groups against as many stones drawn 9,999 times from the same district (within 10 kilometres, outside the group's own site).

4.5 Sensitivity analyses (the addendum plan). Grave fields: raw model; the plan's model; carver as a random effect (Bates et al. 2015); parish as a random effect; location as a spatial surface (Wood 2017); each half; without one province at a time; without stones Runor records as moved. Names: each half; each province alone; without pairs with the same secure carver; without stones moved according to Runor. A step holds when the effect goes the same way with p below 0.05; when a province is removed or tested alone, significance is required only where at least two thirds of the material remains.

4.6 Deviations from the plans. In step 1 the plan's 2-kilometre bandwidth gave stones without neighbours a density near zero, so the test measured nothing, as a calibration on random patterns showed. That run is reported as invalid; an addendum with a 10-kilometre bandwidth and a floor was locked before the new run, and the corrected method was then replicated on new stones (section 2), with only one 19-simulation test run to find code errors. In step 3 provinces with fewer than ten stones were merged. Otherwise none.

5. Results

5.1 The stones form groups. The stones lie closer together than chance would give at 100–1,000 metres: Uppland p = 0.004, Södermanland p = 0.003 (Holm-corrected 0.006 both), in both halves and

with every bandwidth and floor tried (p 0.001–0.005). The median distance to the nearest stone is 1.1 kilometres in Uppland and 1.4 in Södermanland, but one stone in ten has a neighbour within 200 metres. In Götaland the replication agrees: clustered, p = 0.004, in both halves (west p = 0.02, east p = 0.01), and in Västergötland (p = 0.003) and Småland (p = 0.02) alone; Östergötland goes the same way but is too small to be sure (p = 0.20).

5.2 Neighbouring stones share personal names. Of the 74 near pairs where both stones carry names, 13 (17.6 percent) share a personal name, against 2.2 percent of the 4,983 comparison pairs: 7.9 times as often, Holm-corrected p = 0.001. Most near pairs thus share no name. Same carver (2.2 times) and same style group (1.6 times) are indications; bridges, journeys and Christian features make neighbours no more alike. All sensitivity analyses hold:

- Half A: ratio 9.3 (34 near pairs), p = 0.002. Half B: 5.8 (40), p = 0.004.
- Uppland only: 8.2 (63 near pairs, 85 percent of the pairs), p < 0.001. Södermanland only: 5.8 (11 near pairs), p = 0.09 – the same direction, but too few pairs to be sure.
- Without pairs in which both stones have the same secure carver: 7.1, p < 0.001.
- Without stones that Runor records as moved: 9.6, p < 0.001.

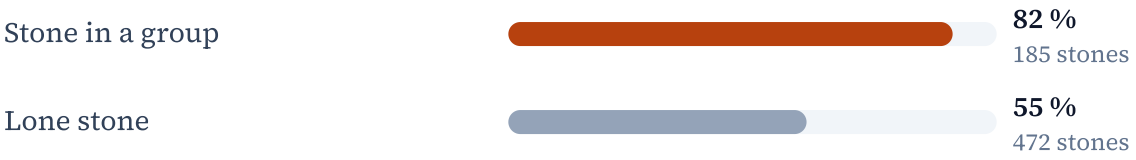
5.3 The groups stand by the grave fields. 82 percent of the stones in a group have a grave field within 500 metres, against 55 percent of the lone ones. With province, dating period and local density constant the odds ratio is 2.38 (1.50–3.78), p < 0.001, Holm-corrected p = 0.003. All sensitivity analyses hold:

- Raw model: odds ratio 3.59. Carver as a random effect (324 groups): 2.49, p < 0.001. Parish as a random effect (331 parishes): 2.45, p < 0.001. Location as a spatial surface: 2.45, p < 0.001.
- Half A: 2.27, p = 0.009. Half B: 2.53, p = 0.01.
- Without stones moved according to Runor (540 stones): 2.64, p < 0.001.
- Without one province at a time: 2.18–2.70 without each of Östergötland, Småland, Södermanland and Västergötland; without Uppland (46 percent of the stones left) 1.44, not significant (p = 0.35): the same direction, but the difference is carried largely by Uppland.

For burial mounds, roads and boundary markers no difference is shown: road within 500 metres 26 against 16 percent raw, odds ratio 1.32 (0.83–2.11); boundary marker 7 against 6 percent. With density held constant, place names in Husby and -tuna lie somewhat *less often* near the groups (odds ratio 0.53, Holm-corrected p = 0.049) and a medieval church within 2 kilometres somewhat more often, 34 against 17 percent (odds ratio 1.66, 1.06–2.60, p = 0.03, Holm-corrected p = 0.34): both indications. Old and young settlement names, cult names, a place in the text, inheritance or farm, height and signing show no demonstrated difference, nor does the share of near pairs with exactly one Christian stone differ clearly from chance (23 against 31 percent, p = 0.10).

Groups against lone stones, and near against distant pairs

Grave field within 500 m



The same personal name



The scale runs from 0 to 90 %. Source: Rundata; grave fields from Kulturmiljöregistret, Swedish National Heritage Board. Raw shares. Chart: Vitki, after Rundata and Kulturmiljöregistret.

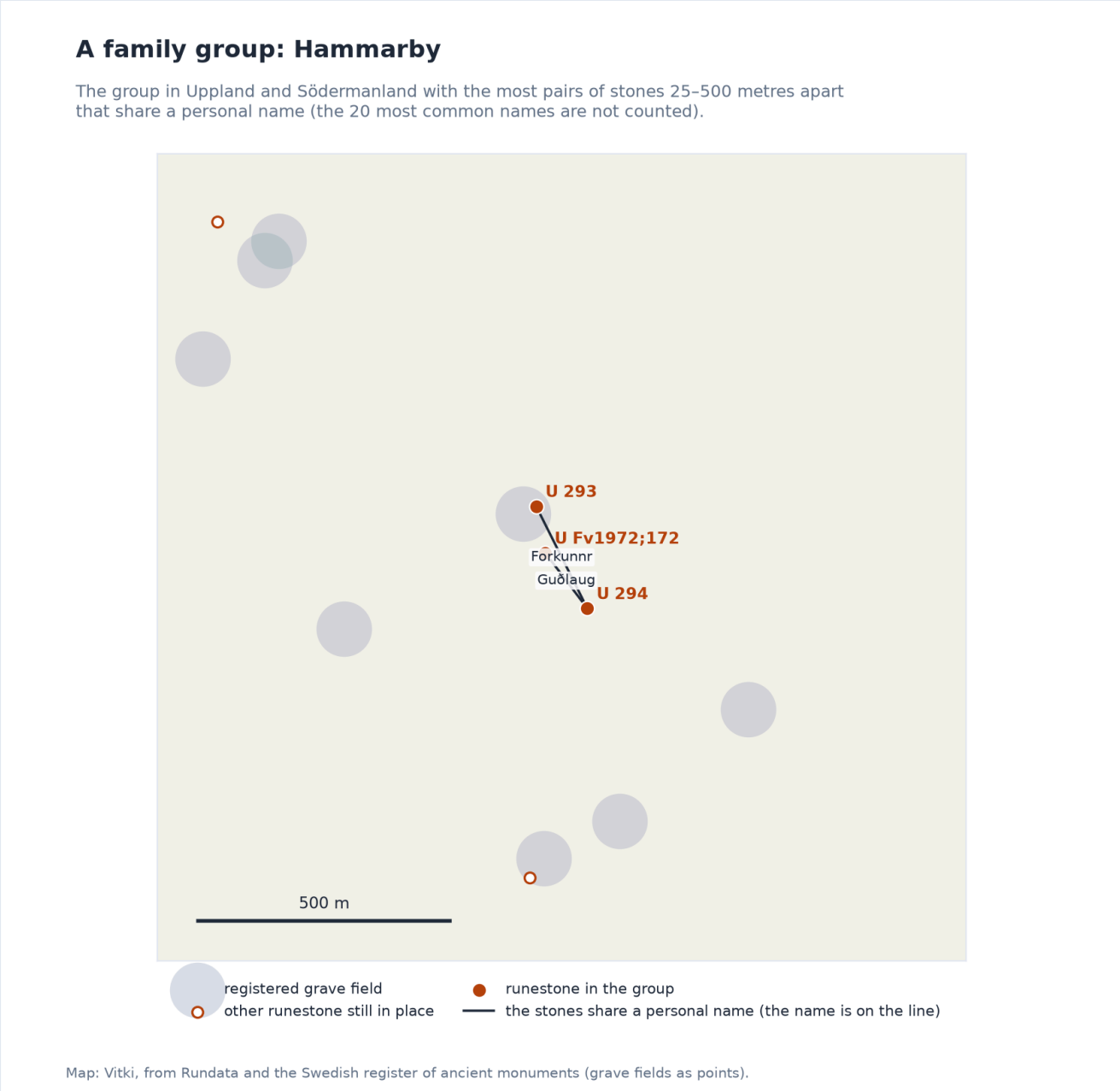


Figure 1. Hammarby parish, Uppland: of the 13 pairs of stones 25–500 metres apart that share a personal name, these form the only group with two pairs (U 293 and U 294 both name Forkunnr; U 294 and U Fv1972;172 both Guðlaug). Grey discs: registered grave fields, at their point. Map: Vitki, after Rundata and Kulturmiljöregistret (Swedish National Heritage Board).

5.4 Follow-up: a group was raised within one generation. It has the plan's own robustness checks, not the fixed series of sensitivity analyses. 36 groups with 84 style-dated stones (32 in Uppland, 4 in Södermanland) have on average 1.58 different style groups, against 1.93 for as many stones from the same district, $p < 0.001$; half A $p = 0.003$, half B $p = 0.004$. 47 percent of the groups have all stones in one style group against 21 percent expected; a group's earliest and latest stones are on average 20 years apart against 36. Without groups where one carver made every stone (28 remain): 1.71 against 1.96, $p = 0.01$, weaker but present. Example: Täby (U 135–137, 140, all Pr2); counter-example: Vårfrukyrka (Fp and Pr4).

In plain words. Runestones stand closer together than chance gives. Stones within 500 metres share a personal name about eight times as often as stones 2–10 kilometres apart. Four in five grouped stones stand by a grave field, just over half of lone stones do. By a follow-up not yet sensitivity-tested, a group's earliest and latest stones lie on average twenty years apart, closer than stones from the same district.

6. Discussion

What the findings say. Four measurements point the same way. The simplest reading is that the group is a family's place, where several stones were raised, by a follow-up not yet sensitivity-tested within one generation; whom the stones commemorate was not measured. Neighbouring stones are not more often both bridge or journey stones, and the difference for roads (5.3) is not clear. The register's roads and boundary markers are moreover mostly younger than the stones.

Against earlier research. The findings support Klos (2009) on the grave field as a main place for runestones, but sharpen it: above all the *groups* stand by grave fields, while just over half of the lone stones do too. Ekholm's (1950) and McLeod's (2014) road stones and Zachrisson's (1998) boundary stones cannot be tested well with the register's younger roads and boundary markers; the comparison gives them no support but does not rule them out. That lone stones less often have a grave field nearby cannot separate Ekholm's reading, that they never stood by one, from the one he argued against, that the grave fields were ploughed away (weakness 3).

Weaknesses, in order of weight.

1. **Uppland carries much.** 32 of 36 groups in the generation test and 85 percent of the near pairs in the name test lie in Uppland, and without Uppland the grave-field difference is not significant; only the clustering is also shown elsewhere.
2. **Only stones in place,** about a third of all. If moved stones came more often from groups or from lone places, comparisons are affected. Names and grave fields hold without the stones Runor records as moved, but step 1, the spacing, has no such check.
3. **The registers are incomplete and younger.** A ploughed-out grave field is not in the register, so a stone may be lone in the register though it once stood by a grave field – the explanation Ekholm argued against. The younger roads and boundary markers leave both poorly tested.
4. **Step 4 rests on the same stones as the observation behind it,** and style groups are 25–50 years wide. The style group often follows the carver or the workshop, so "within one generation" may partly mean "by the same unknown carver".
5. **The names are read by rules,** and a name can be borne by different people; removing the most common names reduces but does not remove that risk.

What would overturn the findings. Shared names that vanished with the carver or the sponsor's formula held constant, groups that followed roads in a register of period roads, or a generation pattern that did not recur in another province with many stones in place.

7. Conclusion

Runestones often stand in groups – in Uppland and Södermanland, and by a pre-registered replication in Götaland – that belong to families: neighbouring stones share personal names eight times as often as distant ones, and groups stand by grave fields more often than lone stones. A follow-up, not yet sensitivity-tested, suggests that a group was usually raised within one generation. This applies first of all to Uppland.

Data and code

The analysis was run in Vitki 3.0.0, an open research tool for runestones (https://vitkirunor.se/en). The result files were saved in https://github.com/Precatio/runes at commit `6b0680c` (the spacing), `56b95a7` (the neighbouring stones), `64f2c7f` (the family's place) and `777ec0a` (one generation), all 2026-10-29, and `cae41e0` (the sensitivity analyses) and `34281c9` (the replication), both 2026-10-31; the analysis was run on the code at those commits. All code, the analysis plans and the result files are openly available there under GPL-3.0.

- The material: corpus `3f58e9339191761e`, Rundata's Swedish Viking-Age runestones as frozen on 2026-10-28 (`data/cache/r/corpus/3f58e9339191761e/corpus.csv`).
- Data from other sources: Ancient monuments: Kulturmiljöregistret, Swedish National Heritage Board (via K-samsök). Place names: © OpenStreetMap contributors (ODbL). Medieval churches: Wikidata (CC0). Moved stones: Runor (2020 edition), retrieved 2026-10-16 (`data/raa/runor_2020.json`).
- The analysis plans: `analysplaner/2026-10-28-avstand-mellan-stenar.md` with its addendum, `analysplaner/2026-10-29-grannstenarna.md`, `analysplaner/2026-10-29-familjens-plats.md`, `analysplaner/2026-10-29-en-generation.md`, `analysplaner/2026-10-31-familjens-plats-trappan.md` and `analysplaner/2026-10-31-avstand-replikering.md`, each with its `.sha256` file.
- The scripts: the spacing `utdata/avstand/avstand.R`, the replication `utdata/avstand/replikering.R`; the neighbouring stones `utdata/avstand/grannar.py` and `utdata/avstand/grannar_trappa.py`; the family's place and the generation `utdata/familjens_plats/data.py`, `analys.R`, `generation.py` and `trappa.R`; results and deviations in `resultat.md`, `generation_resultat.md` and `trappan_resultat.md` in the same folders.
- To rerun the analysis, from the project folder: `Rscript utdata/avstand/avstand.R data/cache/r/corpus/3f58e9339191761e/corpus.csv data/cache/r/naturalearth data/raa/lamningar.csv utdata/avstand`; `Rscript utdata/avstand/replikering.R data/cache/r/corpus/3f58e9339191761e/corpus.csv data/cache/r/naturalearth utdata/avstand`; `.venv/bin/python utdata/avstand/grannar.py`; `.venv/bin/python utdata/avstand/grannar_trappa.py`; `.venv/bin/python utdata/familjens_plats/data.py`; `Rscript utdata/familjens_plats/analys.R`; `.venv/bin/python utdata/familjens_plats/generation.py`; `Rscript utdata/familjens_plats/trappa.R`.
- The report is built by `utdata/blogg/build/build_rapport_familjens_plats.py`, which reads every figure from the result files and stops if anything disagrees (the freeze date and Runor's retrieval

date from `utdata/familjens_plats/rapport_kontroll.json`); the builder and the check file as reviewed before archiving are in commit `6eae0af`.

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Change log

- **Version 2, 2026-11-01.** Figure 1 added: a map of a real family group by grave fields, chosen by a rule given in the caption. No numbers, interpretations or conclusions have changed.
- **Version 3, 2026-11-05.** Corrections before the permanent archiving in Zenodo, after a review of the report against the result files, the plans and the sources. The main results and the level are unchanged.
- *Pre-registered tests that were not reported.* The test of medieval churches (34 against 17 percent, Holm-corrected p = 0.34, an indication) and the tests of settlement names, cult names, a place in

the text, inheritance or farm, the stone's height, signing and Christian and non-Christian neighbours were missing from the results. They are now in 5.3, and section 2 lists all fifteen tests.

- *Roads and boundaries.* The abstract, the discussion and the essentials box said that roads, boundary markers and the place names of power do not set the groups apart and that the monument was the family's, "not the road's or the boundary's". Roads are more common by the groups in raw shares (26 against 16 percent), but the difference is not clear; Husby and -tuna are an indication in the opposite direction; and the register's roads and boundary markers are mostly younger than the stones, a caveat in the plan that was missing. The text now says so.
- *The same personal names, not the same people.* The measure is a shared personal name, and only 13 of 74 near pairs share one. The title and the text now say personal names; the address is unchanged.
- *The carver.* Material gave the carver as "secure or uncertain"; the analyses count only a carver that Rundata gives as secure.
- *The literature.* Ekholm (1950) is now described from his text: the stones stood mainly by roads and waterways, and the lone stones are mostly road monuments, not remnants of ploughed-out grave fields. That lone stones less often have a grave field nearby cannot separate his reading from the one he argued against; the discussion and weakness 3 are corrected. Zachrisson's (1998) stones marked the farm's boundaries, often where the road entered the farm's land. Ekholm and McLeod now have pages.
- *Step 4.* One generation is a follow-up that has not passed the fixed series of sensitivity analyses; this is now said wherever the step is mentioned, and a weakness is added: the style group often follows the carver or the workshop.
- *Sources and form.* The place names (OpenStreetMap) and the churches (Wikidata) now have their sources; Runor is cited as the 2020 edition with its retrieval date; the material was frozen on 2026-10-28, not 2026-10-29; weakness 2 says that step 1 has no check without the stones Runor records as moved; the caption says discs instead of rings; one stone in ten has a neighbour within 200 metres, not "about 200"; Data and code gives the commits and commands instead of a future DOI; Rundata is cited with version and licences; the glossary is linked and the spatial terms and one-sided test are explained; the English text is shortened to at most 3,000 words.
- **Version 4, 2026-11-05.** Corrections after a second, independent reading of the changes in version 3. The main results and the level are unchanged.
- *Figure 1.* The image's own text said that the stones "name the same person". The image is redrawn and now says "share a personal name", as the text does; the map and the group are the same.
- *Step 4.* Version 3 said that the caveat on one generation is given wherever the step is mentioned, but it was missing in "In plain words" and in the discussion. It is now there too.
- *The discussion (6).* "For a father, sons and brothers" is struck: whom the stones in a group commemorate was not measured. The figures for roads are in 5.3, and the discussion refers there.
- *The lede and the essentials box.* "The runestone was the family's monument by the family's grave field" stood as a finding; it now stands as a reading that the results fit.
- *Zachrisson (1998).* Version 3 spelt the title "tidigmedeltid" and called it a correction. The dissertation catalogue (avhandlingar.se) and the project's literature register have "tidig medeltid", which is restored; the DiVA record could not be reached. The claim about the spelling is struck from version 3's entry.

- *Data and code.* The builder and the check file as reviewed before archiving are named with commit `6eae0af`.
- *What version 3 also changed without logging it.* Four sentences were removed: that one stone per farm, evenly spread, finds no support (5.1), the Skokloster example (5.4), that the generation pattern had not been tested statistically before (6), and the next step (7).

Competing interests: none.

Funding: none.

Cite as

Kvant, V. (2026). Runestones gather in family groups by the grave fields – neighbouring stones name the same personal names eight times as often. Research report, report version 4; analysis in Vitki 3.0.0. https://vitkirunor.se/en/forskningsrapporter/familjegrupper-vid-gravfalten